

Java Project Task Documentation

Project Title: Smart Console-Based Expense Tracker (Core Java Only)

Objective:

Build a fully functional **console-based expense tracker** using **Core Java only**, implementing your **own data structures** (LinkedList, HashMap, etc.), and providing **report generation**, **data persistence**, and **category-wise spending analysis**. This project simulates a real-world application and tests your understanding of Java fundamentals and problem-solving ability.

Constraints:

- **NO Java Collections** like `ArrayList`, `HashMap`, `LinkedList`, etc.
 - **NO third-party libraries** (no Spring, Hibernate, Lombok, etc.)
 - **NO GUI** — strictly **console-based**
 - You **must write your own implementations** of required data structures (e.g., `MyLinkedList`, `MyHashMap`)
 - The system should **persist data to a file** (`.txt` or `.csv`) so data is not lost after restart
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Key Features (All Mandatory)

1. User Registration & Login

- Register and login using username/password
- Validate existing users from a file (persist user data)
- Hash the passwords using a simple hashing algorithm (your own — not Java's `MessageDigest`)

2. Add Expenses

- Each user can add an expense with:
 - Amount
 - Category (user-defined or default: Food, Rent, Travel, etc.)
 - Description
 - Date (use Java `Date` or `Calendar`)
- Store these entries in your own data structure like `MyLinkedList`

3. Generate Monthly Summary

- Print a month-wise summary:
 - Total expense per category
 - Total monthly expense
 - Highest expense and its category
- Store reports as `.txt` or `.csv` with readable formatting

4. Category Insights

- Track and show:
 - Most used category
 - Average spending per category
 - Custom alerts if monthly total exceeds a hardcoded limit (e.g., ₹20,000)

5. Data Persistence

- Expenses and users should persist in `.txt` or `.csv` files.
- On login, users should see previous data.

Folder Structure Guidelines (Expected)

pgsql

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ExpenseTracker/

```
|
|— models/
|   |— Expense.java
|   |— User.java
|
|— datastructures/
|   |— MyLinkedList.java
|   |— MyHashMap.java
|
|— utils/
|   |— FileUtil.java
|   |— InputValidator.java
|
|— service/
|   |— ExpenseService.java
|   |— ReportService.java
|
|— main/
|   |— MainApp.java
|
|— reports/
|   |— (Generated Reports)
```

Requirements to Submit

1.  Source code with working folder structure
2.  A short **video (3–5 mins)** explaining how it works, screen-recorded only
3.  A **README.md or PDF (1–2 pages)** with:

- Your approach
 - Custom data structures explanation
 - Sample input/output
4. ☒ At least one **generated .csv or .txt report**
 5. ☒ Manual test cases used to verify outputs

 Bonus (Optional, for brownie points):

- Track recurring expenses
- Build a savings goal feature
- Add sorting (by amount, date) using your own sorting logic